Diagnosing Autism Spectrum Disorder from 3D MRI Images in Pediatric Cases

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Abstract

Autism Spectrum Disorder (ASD) presents a diagnostic challenge due to its heterogeneous nature. Current diagnostic practices, involving extensive behavioral testing by professionals, can be subjective and time-consuming. Emerging research suggests that fMRI scans could detect abnormal neuronal connectivity linked to ASD in children. This study explores the efficacy of various machine learning (ML) and deep learning (DL) models in distinguishing between fMRIs of ASD and non-ASD patients. Baseline models such as logistic regression, random forest, and support vector machines were evaluated alongside 2D CNNs, pre-trained 2D CNNs, 2D convolutional autoencoders, and 3D CNNs. Surprisingly, the random forest model showed the best accuracy (64%), while the highest performing deep learning model—a 2D CNN with a Gaussian Noise filter—achieved 62.7% accuracy. The study underscores the need for further exploration of data augmentation methods and improved machine learning models, especially given the constraints of dataset size and computational resources.

Introduction & Background

Autism Spectrum Disorder (ASD) is a multifaceted, pervasive, and complex disorder that affects 1 in 36 children in the US [1]. The broad symptomatology of ASD in children contributes to diagnostic complexity due to its high heterogeneity, making accurate diagnosis challenging [2, 3]. Currently, there is no formal medical test for pediatric ASD as diagnosis is determined based on clinical observations by trained medical professionals who evaluate socio-environmental factors that implicate the disease, such as a patient's behavioral dispositions, social interaction, cognition, communication, developmental milestones, among others [4]. Given the growing evidence that suggests that autism is the by-product of changes in brain organization that manifests as abnormal neuronal connectivity [5,6]; the primary research question that this project is concerned with is whether a deep learning (DL) model can be used to effectively differentiate ASD and non-ASD pediatric patients using MRI scans. The primary research question in this study is whether a deep learning (DL) model can be used to effectively differentiate ASD and non-ASD pediatric patients from MRI scans.

Methods - Data Processing

Dataset: This study utilized neuroimaging data from the Autism Brain Imaging Data Exchange (ABIDE), which includes imaging data from 539 individuals with ASD and 573 typically developing controls (TDC), collected from 17 sites [1, 5]. We focused on ABIDE-I data for its extensive quality control, which involved manual checks by three independent reviewers. As observed in **Figure 1**, the dataset primarily includes children under 18, with significantly more males than females, reflecting the higher prevalence of ASD among males [7].

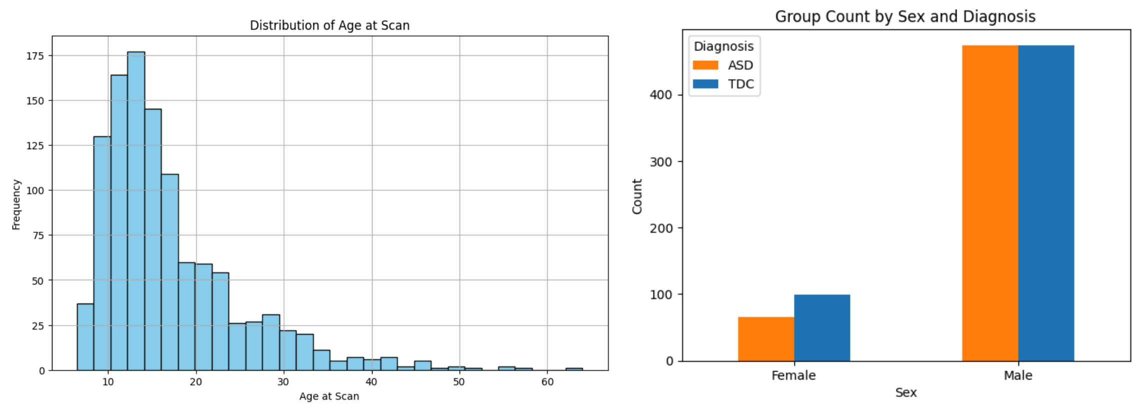

Figure 1. ABIDE-I Data Distribution. *This figure shows the distribution of the 1112 patients found in the ABIDE-1 dataset. Patients are aged between 6 - 64 years of age, with the majority being children < 18 years of age. There are significantly more male cases than female cases in the dataset, but the stratification between ASD and TDC are reasonably equal between classes.*

Preprocessing fMRI Data: We used preprocessed neuroimaging data from the Preprocessed Connectomes Project (PCP) via ABIDE-I, employing the Configurable Pipeline for the Analysis of Connectomes (CPAC) for its comprehensive preprocessing steps. These included motion correction, global mean intensity normalization, and functional data standardization to MNI space (3x3x3 mm resolution), followed by band-pass filtering (0.01-0.1Hz) and global signal regression to reduce noise.

Feature Selection: To characterize brain connectivity, we used Degree Centrality weighted by correlation, which quantifies the number and strength of connections for each brain region. This metric offered simplified representation, network insights, and noise reduction, providing a focused understanding of key network features associated with ASD. This allowed the classifier to better distinguish ASD from control subjects using streamlined input features.

***Cohort Selection*:** We filtered the dataset to include only data with valid labels, from patients aged 18 or younger, and with consensus among the three reviewers on data quality. This age filter was crucial as ASD symptoms typically manifest in early childhood, with presentations in adults differing from those in children. After filtering, we retained 614 cases: 505 males (262 ASD, 243 TDC) and 109 females (38 ASD, 71 TDC).

***Stratified Sampling*:** To balance the dataset for ASD and TDC patients while maintaining similar gender distributions, we used stratified sampling. This approach was critical given the significant gender disparity in ASD prevalence, allowing us to maintain balanced training and testing sets while preserving this imbalance. Sex and diagnosis labels were encoded (Male = 1, Female = 2) and (TDC = 0, ASD = 1) and combined into tuples for each patient, capturing the dual stratification criteria. Using scikit-learn's StratifiedShuffleSplit, we split the data to ensure representative distributions in both train and test sets, with 80% of the data for training and 20% for testing; the distribution of which can be observed in **Figure 2**.

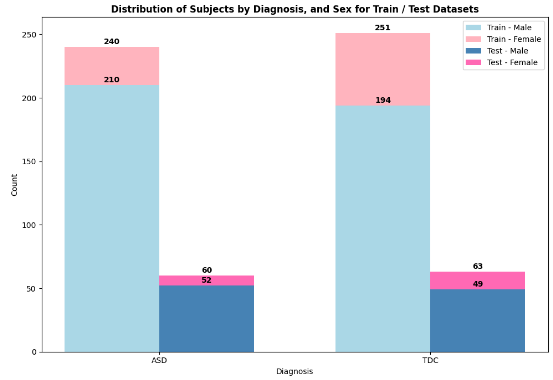

Figure 2. Data Distribution. *Distribution of the data after stratified sampling for ASD and TDC cases across training and testing sets. The training data included 491 patients (87 female, 30 ASD, 57 TDC; 404 male, 210 ASD, 194 TDC), while the testing data included 123 patients (22 female, 8 ASD, 14 TDC; 101 male, 52 ASD, 49 TDC).*

***Data Loading and Resizing*:** The original fMRI images had dimensions of 61x73x61. To standardize input size and facilitate efficient deep learning processing, we resized each image to 64×64×64 using cubic interpolation with the *ndimage.zoom* function from the SciPy library. By calculating appropriate zoom factors based on the original dimensions, each image was resized to the target size, maintaining the original aspect ratio and spatial relationships. This strategy provided a standardized image format compatible with convolutional neural networks (CNNs) like U-Net, which benefit from even dimensions to align upsampling and downsampling layers. Standardized input dimensions also simplified model architectures and improved accuracy, as models trained on uniformly-sized data tended to perform better. Images were read from the local directory using the nibabel library (nib.load), converting the data to NumPy arrays in np.float32 format for numerical consistency.

Data Transformation: To standardize all images for deep learning and visualization, we implemented a custom transformation function, transform_and_reorder_mri_dataset. This function reoriented the axes and reordered dimensions to ensure a consistent orientation across all samples. The original data had dimensions ordered as X (sagittal), Y (coronal), and Z (axial). The transformation rearranged these to Z-Y-X and applied specific rotations to standardize the orientation: (1) Axial slices, originally on the Z-axis, were rotated 90 degrees counterclockwise and shifted to the X-axis for correct top-down orientation (2) Coronal slices, initially on the Y-axis, were rotated 90 degrees counterclockwise to align the front-to-back orientation, remaining in the Y-axis. (3) Sagittal slices, originally on the X-axis, were rotated 90 degrees counterclockwise, horizontally flipped for standardized left-right orientation, and shifted to the Z-axis. This transformation ensured a standardized dataset, suitable for deep learning and accurate data visualization, as seen in **Figure 3**.

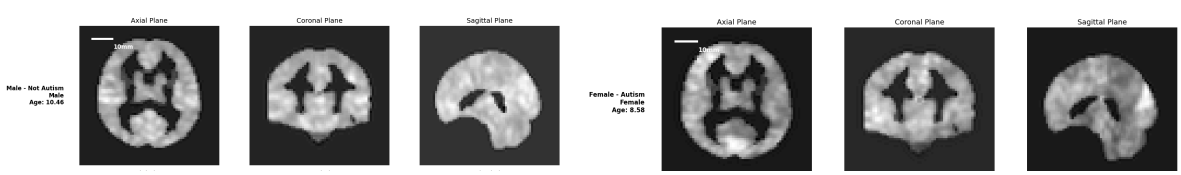

Figure 3. Visualization of Pre-processed dataset. *Left image shows a male without autism aged 10, and the right image shows a female with autism aged 9, across the three axial, coronal, and sagittal planes.*

Data Augmentation: Given the limited size of the dataset and the complex patterns in the fMRI data, we implemented a comprehensive data augmentation pipeline using Keras layers to enhance the generalization and robustness of our deep learning models. To enrich the training data, various transformations were applied. Random flipping ensured the model became invariant to slight positional differences, while random rotation, set at a factor of 0.1, enabled the model to handle minor misalignments. The use of random zoom, also at 0.1, helped the model achieve scale invariance by recognizing features from varying perspectives. Random translation introduced positional shifts to prepare the model for translational variances in real-world data. Finally, Gaussian noise with a standard deviation of 0.1 simulated inherent noise in MRI data, encouraging the model to focus on significant features. The effects of these transformations can be seen in **Figure 4**.

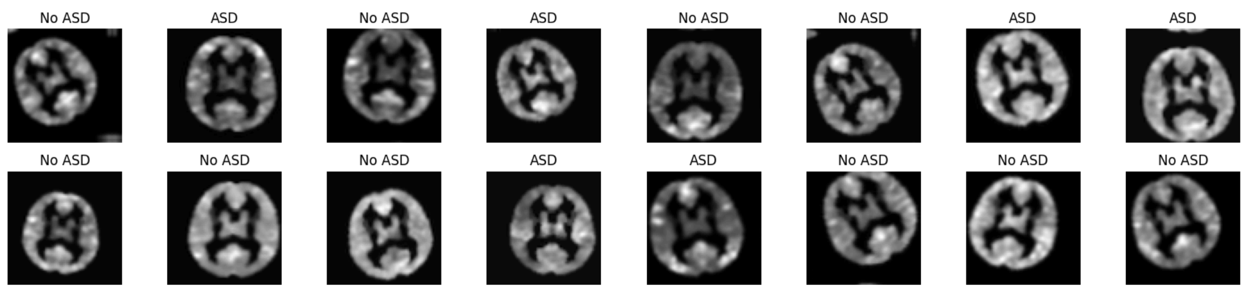

Figure 4. Visualization of the Effect of Data Augmentation. *This figure shows the effect of applying data augmentation on one of the Axial slices found within the first batch of 32 patients.*

Methods - Deep Learning Methods

In this study, we explored four architectural categories: (1) simple machine learning classifiers (Random Forest, Logistic Regression, and SVM) as a baseline, (2) 2D convolutional neural networks (CNNs), (3) 3D CNNs, and (4) unsupervised representation learning using convolutional autoencoders. This approach, layered from simple to more complex models, allowed us to assess the progression in performance and understand the impact of model complexity on classification accuracy. Each category had unique data processing considerations, given that the dataset was 3D and different architectures had varying input shape requirements. The following outlines the architectural categories and their specific approaches.

Simple Machine Learning Classifiers: As a baseline, we used Random Forest, Logistic Regression, and SVM models to explore the benefits and drawbacks compared to more complex deep learning methods. These models are computationally less demanding and offer a quick estimate of classification performance using traditional approaches. The 3D data (64x64x64) was flattened into a 1D vector (262,144) to serve as input for these models. Model configurations are provided in Table 1.

Model	Configuration
Random Forest	Baseline
Random Forest	max_depth = 4 n_estimators = 600
Logistic Regression	Solver = lbfgs
Logistic Regression	Solver = liblinear
Logistic Regression	Solver = liblinear, penalty = l1
SVM	Kernel = linear
SVM	Kernel = rbf

Table 1. *The configurations of simple machine learning models used to predict ASD on MRI scans.*

2D Convolutional Neural Networks (2D CNN): 2D CNNs were chosen for their ability to learn hierarchical features directly from images without manual feature extraction. They excel in handling 2D image data and can capture complex spatial patterns with their layered structure. This approach enabled us to assess how deep learning might enhance classification performance compared to traditional methods while maintaining moderate computational requirements. The challenge in dealing with volumetric data lay in converting the input into a format suitable for 2D CNNs. We explored several methods: (1) Extracting 2D slices from all samples, using either a single plane (axial, coronal, or sagittal) or stacking slices from multiple planes to capture multiple views; (2) Selecting a single representative slice, such as the middle slice, for each sample; (3) Averaging all slices from one view; (4) Using MaxPooling and GlobalPooling to flatten the data; (5) Reshaping the data from 64x64x64 to 512x512. All methods had significant limitations due to information loss. This paper reports the results from using the Average Pooling / MaxPooling transformation. The configurations of our 2D models are presented in Table 2.

Model	Label	Configuration
2D CNN	**V1**	A 2D convolutional neural network (CNN) with five Conv2D layers (64 filters for the first three layers and 128 filters for the last two layers, kernel size 2, activation ReLU, padding 'same'), each followed by MaxPooling2D and Dropout layers for regularization, ending with a Flatten layer, a Dense layer with 64 units (ReLU), and a final Dense output layer with sigmoid activation for binary classification.
2D CNN	**V2**	A 2D convolutional neural network (CNN) starting with a GaussianNoise layer (standard deviation 0.1), followed by five Conv2D layers (64 filters for the first three layers and 128 filters for the last two layers, kernel size 2, activation ReLU, padding 'same'), with each layer being followed by MaxPooling2D and Dropout for regularization, finishing with a Flatten layer, a Dense layer with 64 units (ReLU), and a final Dense output layer with sigmoid activation for binary classification.
ResNet50	**V1**	A hybrid deep learning architecture that starts with three Conv3D layers (64, 16 filters, and a Conv3DTranspose layer, kernel sizes 3 and 2, activation ReLU) to process 3D input arrays, followed by reshaping to 2D in order to feed into a base ResNet50 model, which includes GlobalAveragePooling2D, two Dense layers (128 and 32 units, ReLU, Dropout 0.25), and a final Dense output layer with sigmoid activation for binary classification.
DenseNet121	**V1**	As above, except the base model is replaced by a DenseNet121 model.
DenseNet121	**V2**	A deep learning architecture utilizing DenseNet121 as the base model (with layers frozen during training), followed by a GlobalAveragePooling2D layer, a Dense layer with 64 units (ReLU), and a Dense output layer with sigmoid activation for binary classification, optimized using the Adam optimizer with a learning rate of 0.001 and compiled with binary cross-entropy loss.

Table 2. *The configurations of 2D CNN architectures to predict ASD on MRI scans.*

2D Convolutional Autoencoders (2D CAE): We investigated convolutional autoencoders to determine if their unsupervised representation learning capabilities could learn a compressed representation of the input data. By encoding images into a lower-dimensional latent space and then reconstructing them, this compression process aimed to help the model learn intrinsic patterns in the data. These nuanced features, captured in the latent space, could potentially improve the downstream ASD classification task by leveraging patterns not easily detectable through traditional supervised learning methods. The configuration used can be seen in **Table 3.**

Model	Label	Configuration
2D CAE	**V1**	A 2D convolutional autoencoder with an encoder consisting of three Conv2D layers (128, 64, and 32 filters, kernel size 2, activation ReLU) and two MaxPooling2D layers for downsampling, followed by a decoder with two UpSampling2D layers and Conv2D layers, culminating in a final Conv2D output layer with sigmoid activation, optimized with a custom KL divergence loss for unsupervised representation learning.

Table 3. *The configuration of a 2D CAE used for representation learning with a downstream classifier.*

3D Convolutional Neural Networks (3D CNN): We extended the 2D CNN architectures to 3D CNNs, which are better suited for three-dimensional image data. For fMRI data, this architecture allows the model to capture the full volumetric information in the scans, offering richer spatial context. The added complexity and computational cost of 3D CNNs is justified by their potential to identify intricate patterns across the entire brain, significantly enhancing classification performance. The data transformation step involved expanding the 3D volumetric data to include a fourth dimension, resulting in a shape of (64, 64, 64, 1) for grayscale images.

Model	Label	Configuration
3D CNN	**V1**	A 3D CNN with four Conv3D layers (32, 64, 128, and 256 filters, kernel size 3), each followed by MaxPool3D, BatchNormalization, and ReLU activation; ending with a GlobalAveragePooling3D layer, a Dense layer with 256 units (ReLU, dropout 0.5), and a final Dense output layer with a sigmoid activation.
3D CNN	**V2**	A 3D CNN with three Conv3D layers (32, 64, and 128 filters, kernel size 3, activation ELU), each followed by MaxPooling3D, culminating in Flatten, two Dense layers with 64 units (ELU), and a final Dense output layer with a sigmoid activation, incorporating dropout at 0.5 for regularization.
3D UNet	**V1**	A 3D U-Net with an encoder consisting of three Conv3D layers (64, 128, and 256 filters, kernel size 3, activation ReLU) each followed by BatchNormalization and MaxPooling3D, and a bottleneck Conv3D layer (512 filters, kernel size 3, activation ReLU, BatchNormalization). The decoder includes three Conv3DTranspose layers (256, 128, and 64 filters) with skip connections to respective encoder layers, concatenated with Conv3D layers (activation ReLU, BatchNormalization), followed by GlobalAveragePooling3D, a Dense layer with 256 units (ReLU, dropout 0.5), and a final Dense output layer with sigmoid activation.
3D UNet	**V2**	A modified 3D U-Net architecture with an encoder of four Conv3D layers (64, 128, 256, and 512 filters), a bottleneck with 1024 filters, all followed by BatchNormalization and MaxPooling3D, a decoder with three Conv3DTranspose layers (512, 256, and 128 filters), and skip connections between corresponding encoder and decoder layers, culminating in GlobalAveragePooling3D, two Dense layers, and a final Dense output layer with sigmoid activation for binary classification.

Table 4. *The configurations of 3D CNN architectures to predict ASD on MRI scans.*

Model Training: The training configuration for all models was standardized to optimize learning efficiency and prevent overfitting. An ExponentialDecay learning rate schedule was used, starting at 0.01 and decaying by 10% every 10,000 steps (decay_rate 0.9, decay_steps 10000), gradually reducing the learning rate as training progressed. The models were compiled with the Adam optimizer to leverage this schedule for efficient convergence, and binary cross-entropy was used as the loss function. Accuracy and AUC (Area Under the Curve) metrics were used to evaluate classification performance. To further enhance training, we utilized two callbacks. Early stopping monitored validation loss (val_loss) and halted training if no improvement was seen in 10 consecutive epochs, restoring the best weights to prevent overfitting. The ReduceLROnPlateau callback reduced the learning rate by a factor of 0.2 if validation loss plateaued for 5 epochs, with a minimum rate of 0.0001. Training was conducted with batch sizes of 32 or 64, and up to 100 epochs with early stopping to efficiently manage training time.

Model Evaluation Metrics: Model training and validation accuracy were assessed using several key metrics. Precision measured the proportion of correctly predicted positives, while recall evaluated the model's ability to identify actual positives. Specificity gauged the accuracy in identifying negatives, and accuracy offered an overall measure of correctness. ROC-AUC assessed the model's classification ability by measuring the area under the receiver operating characteristic curve, capturing the balance between true and false positives.

Results

Baseline ML Models: The results of the simple machine learning models, as presented in **Table 5**, show a range of performances across different model types. In the random forest classifier category, the model with a depth of 4 and 600 estimators performed best, achieving an accuracy of 61.79% and an ROC-AUC of 64.1%, surpassing the baseline. For logistic regression models, the one using the lbfgs solver with 1000 iterations achieved the highest accuracy of 62.6% and an ROC-AUC of 64.26%, proving to be the top performer in its category. Among the SVM models, the linear kernel model outperformed the RBF kernel model, with an accuracy of 60.98% and an ROC-AUC of 62.82%. Overall, the logistic regression model with the lbfgs solver provided the best performance among all simple machine learning models, while the SVM with the RBF kernel showed the lowest accuracy and ROC-AUC, indicating that simpler models can be more effective than others for this classification task.

Model	**Random Forest**	**Random Forest**	**Logistic Reg.**	**Logistic Reg.**	**Logistic Reg.**	**SVM**	**SVM**
Config	baseline	Depth=4, Trees=600	Solver = lbfgs, iter=1000	Solver = lib_linear, iter=1000	Solver= lib_linear, penalty=l1, iter=1000	Kernel = linear	Kernel = rbf
Precision	59.18	63.83	61.67	60	61.29	60	54.84
Recall	48.33	50	61.67	60	63.33	60	28.33
Specificity	68.25	73.02	63.49	61.9	61.9	61.9	77.78
Accuracy	58.54	61.79	62.6	60.98	62.6	60.98	53.66
ROC AUC	62.67	64.1	64.26	62.46	64.02	62.82	54.6

Table 5. *The results of simple machine learning models used to predict ASD on MRI scans.*

2D CNN Models: Among the 2D deep learning models, the 2D CNN model with the GaussianNoise layer (2D CNN V2) demonstrated the best performance, achieving a testing accuracy of 62.7% and an AUC-ROC of 62.165%. The pre-trained transfer learning models (ResNet50 and DenseNet121, in both versions) performed poorly, with AUC-ROC scores around 50% and accuracies hovering around 55%. These results indicate that the 2D CNN architecture specifically designed for this task outperformed the transfer learning models, which struggled to effectively distinguish between classes. The results for these models are summarized in **Table 6**.

Model	2D CNN V1	2D CNN V2	ResNet50 V1	DenseNet121 V1	DenseNet121 V2
Accuracy	57	62.7	54.4	55.9	55.3
AUC-ROC	62.1	62.165	43.4	49.9	50.4

Table 6. *The results of 2D convolutional neural networks used to predict ASD on MRI scans.*

2D CAE Model: The results indicate that the 2D convolutional autoencoder, optimized with mean squared error (MSE) loss, performed reasonably well in reconstruction, achieving an accuracy of 73%. However, its representation did not translate effectively to the downstream task, as evidenced by the dense classification model's reduced accuracy of 57%. This suggests that while the autoencoder may have reconstructed the input data, the learned features may not be well-suited for the classification task.

Model	Convolutional Autoencoder with MSE Loss	Dense Classification with Encoder Input
Accuracy	73	57
Loss	2083	0.69

Table 7. *The results of 3D convolutional neural networks used to predict ASD on MRI scans.*

3D CNN Models: Among the 3D convolutional neural networks (CNNs) evaluated, the 3D CNN V1 model demonstrated the best performance, achieving an accuracy of 54.47% and an ROC AUC of 56.9%. Despite this, all the 3D models underperformed, indicating difficulties in generalizing to this dataset. The potential reasons for this subpar performance could be attributed to factors such as model complexity, insufficient training data, or limitations in capturing meaningful features from the 3D data. Further analysis in the discussion section aims to identify the underlying causes of these results and propose improvements.

Model	CNN 3D V1	CNN 3D V2	UNET 3D V1	UNET 3D V2
Precision	56.67	50	46.9	0
Recall	28.33	3.33	88.33	0
Sensitivity	79.37	96.83	4.76	100
Accuracy	54.47	51.22	45.53	51.22
ROC AUC	56.9	50.05	45.16	45.21

Table 8. *The results of 3D convolutional neural networks used to predict ASD on MRI scans.*

Overall Performance of Models: The performance of the deep learning models across different categories reveals a mixed picture of their effectiveness in classification tasks. Among the 2D models, the custom-designed 2D CNN V2, incorporating a GaussianNoise layer, demonstrated superior performance with an accuracy of 62.7% and an ROC AUC of 62.165%, outperforming transfer learning models such as ResNet50 and DenseNet121, which failed to surpass 55% accuracy. However, the 3D CNN models generally underperformed, with 3D CNN V1 achieving a modest accuracy of 54.47% and an ROC AUC of 56.9%. The 2D convolutional autoencoder, while achieving a strong reconstruction accuracy of 73%, did not translate its representations well into classification tasks, as evidenced by the dense classification model's accuracy of 57%. Overall, these results suggest that for this particular classification task, custom-designed 2D CNN architectures may offer better performance compared to more complex or transfer learning-based 3D architectures.

Comparison with Non-DL Methods: Comparing the performance of deep learning models with non-deep learning approaches reveals nuanced insights into the effectiveness of each method for this task. While the custom-designed 2D CNN V2 model excelled among the deep learning architectures, achieving an accuracy of 62.7% and an ROC AUC of 62.165%, its performance was only marginally better than the best non-deep learning model, which was a logistic regression model with an lbfgs solver, achieving an accuracy of 62.6% and an ROC AUC of 64.26%. This demonstrates that, despite their potential for capturing complex patterns in data, deep learning models may not always significantly outperform simpler models like logistic regression. In this case, careful feature selection and model design may be as important as the choice of model architecture, with simpler non-deep learning models providing competitive, if not superior, performance on this task.

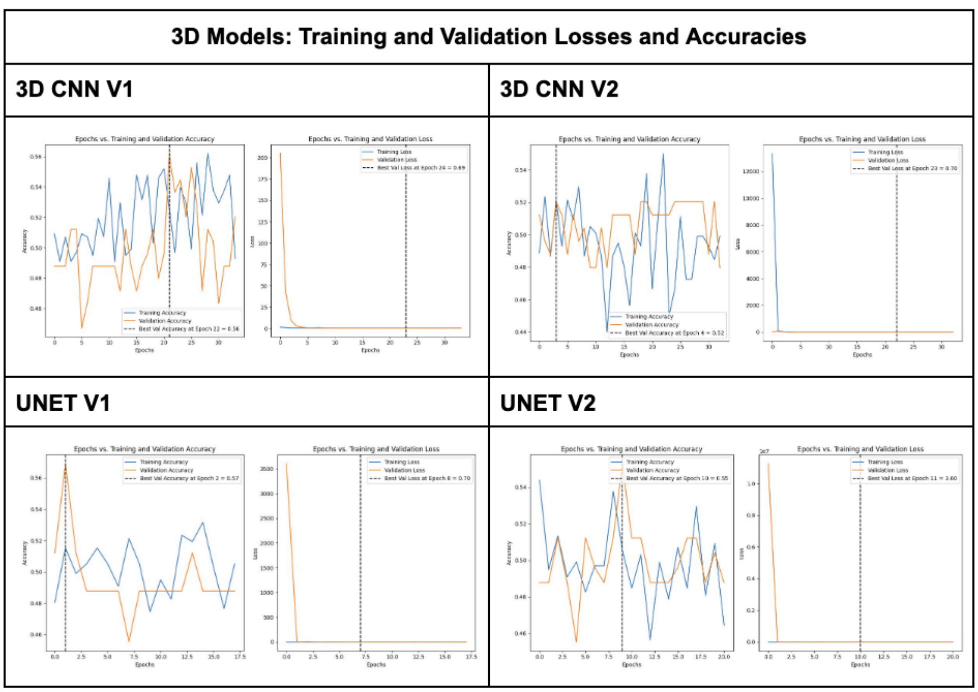

Figure 5: Training and Validation Losses and Accuracies for the 3D CNN Models.

Discussion

Implications of Results: The results of this study reveal several important interpretations and implications regarding the use of deep learning and non-deep learning models in the classification of neuroimaging data. While custom-designed 2D CNN architectures showed marginally better performance compared to traditional machine learning approaches, the differences were not as pronounced as expected. This suggests that the added complexity of deep learning models does not always guarantee significantly improved performance, particularly when the dataset is limited in size or when the task requires subtle pattern recognition. The relatively close performance between the 2D CNN model and logistic regression indicates that feature engineering and model architecture can be crucial determinants of success. Additionally, the poor performance of 3D models underscores the challenges in extracting meaningful features from high-dimensional data, implying that further research is needed to develop more robust 3D architectures and preprocessing techniques. Overall, these findings suggest that while deep learning holds promise for neuroimaging analysis, careful model selection and data processing remain key for achieving optimal results.

Limitations of the study: A significant limitation of this study is the relatively small dataset size, which limited the ability of deep learning models to effectively learn and generalize. With only 614 images available for training and validation, the dataset lacked the scale required for robust training, especially for deep convolutional neural networks (CNNs). This constraint exacerbated the vanishing gradient problem, where gradients diminish significantly as they backpropagate through multiple layers, further impeding weight updates and learning. Consequently, the models likely struggled to extract meaningful features, limiting their generalization capacity. One potential solution is transfer learning, where models are pre-trained on larger MRI datasets to learn robust features, which can then be fine-tuned on the specific dataset in this study. This strategy could mitigate the limited dataset size and enhance model performance. Furthermore, the dataset comprised images from 16 different sites, introducing technical variations that could negatively impact model performance. Variations in acquisition parameters and hardware across sites might have contributed to inconsistent data quality, affecting model performance. Future work should investigate these technical effects and consider pre-processing strategies like batch effect removal to reduce their influence.

Additionally, the choice of input data features presented another limitation. While this study focused on image data, the ABIDE dataset also includes time series data from region-of-interest parcellations that could provide complementary information. Incorporating these features could enable more comprehensive models and improve classification accuracy. Future research should leverage the full range of features in the dataset to better capture the nuances of brain activity patterns associated with autism. Furthermore, there is a high degree of heterogeneity in the manifestation of ASD that changes with age, even over relatively short periods, making it challenging to capture universal features of ASD as children grow [8]. This variability complicates the creation of accurate predictive models, as the neurodevelopmental changes associated with aging can significantly alter the presentation of ASD. Consequently, future research should consider stratifying datasets by age or longitudinally studying the same subjects over time to account for age-related differences in ASD manifestation. This would enable the development of models better suited to the intricacies of ASD presentation across different age groups, potentially leading to more personalized and accurate predictions.

Challenges you have faced: Working with 3D images was a new experience for most members of the team, posing several difficulties in determining the best ways to process the data and design a complex 3D model. One significant challenge we encountered was the lack of computational resources needed to handle 3D data beyond a simple 3D CNN. To address this, we experimented with different methods of manipulating the data to be appropriately formatted for 2D models. However, this led to its own set of limitations, primarily due to the significant loss of information when reducing the dimensionality of the data. In general, one of the most significant challenges we faced was the poor performance across various models. Despite exploring a wide range of 2D and 3D models, we struggled to achieve accuracy levels exceeding 70%. This was indicative of the complexities inherent in the dataset, the limitations of the models used, or the preprocessing strategies employed. We also experimented with transfer learning but encountered further difficulties. The input size required for the models often needed to be 2-3 times the original size, which was computationally expensive for volumetric data. Moreover, the transformations required for resizing frequently obfuscated the data, making it challenging to retain relevant features and impacting the model's ability to learn effectively and thus these results are not presented here.

Conclusion and Future Work

This task requires re-evaluation with a larger, more diverse dataset and better computational resources for thorough analysis. Predicting ASD based on brain MRI images remains a challenging task that necessitates further research, particularly in developing robust 3D architectures and advanced preprocessing techniques for complex neuroimaging data. Improved computational power will enable more sophisticated models, such as self-supervised learning or multi-modal approaches, which could handle the inherent complexity of the data better. Future models reaching accuracy greater than 70% should employ explainability methods to identify critical brain regions influencing ASD prediction. Our initial project proposal included explainability models, but due to poor model performance and limited project time, this was deprioritized. However, explainability is crucial for understanding the intricate brain networks associated with ASD beyond general regions like the frontotemporal lobe and hippocampus. Techniques such as GradCAM, occlusion sensitivity, and saliency maps could be instrumental in revealing key areas of activation in individuals with ASD. Additionally, exploring transfer learning from models pre-trained on similar tasks and using sophisticated data augmentation strategies can increase data variability and enhance model performance. Future studies should also involve collaboration with neuroimaging experts to refine preprocessing methods and better interpret clinical significance, ensuring model predictions align with neurodevelopmental patterns of ASD.

Group Member Contributions

All group members were able to contribute to all parts of the projects through a shared google drive containing the project milestones, a shared GitHub repository for the code, and a shared Canva for the poster.

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